Powering the UTWind VAWT

Developed By: Muhammad Abbas, Paul Lou, Madina Nowrozi, Mohid Tahir
Client: UTWind - Power Systems Subteam
Supervisor: Professor Eric Diller

Project Background

Commercial generators are optimized for high speeds, which limits **Vertical Axis Wind Turbine (VAWT)** performance through **high starting torque** and **reduced efficiency at 100-200 RPM.** This work develops a **custom generator** for the **UTWind 2026 turbine**, optimized for **low starting torque** and **improved efficiency at low RPM**. It establishes a **proof-of-concept** for future in-house generator development.

Key Term:
Starting Torque: Minimum torque to overcome startup resistance

Design Methodology

1 ***Brainstorming***
- Four architectures were evaluated.

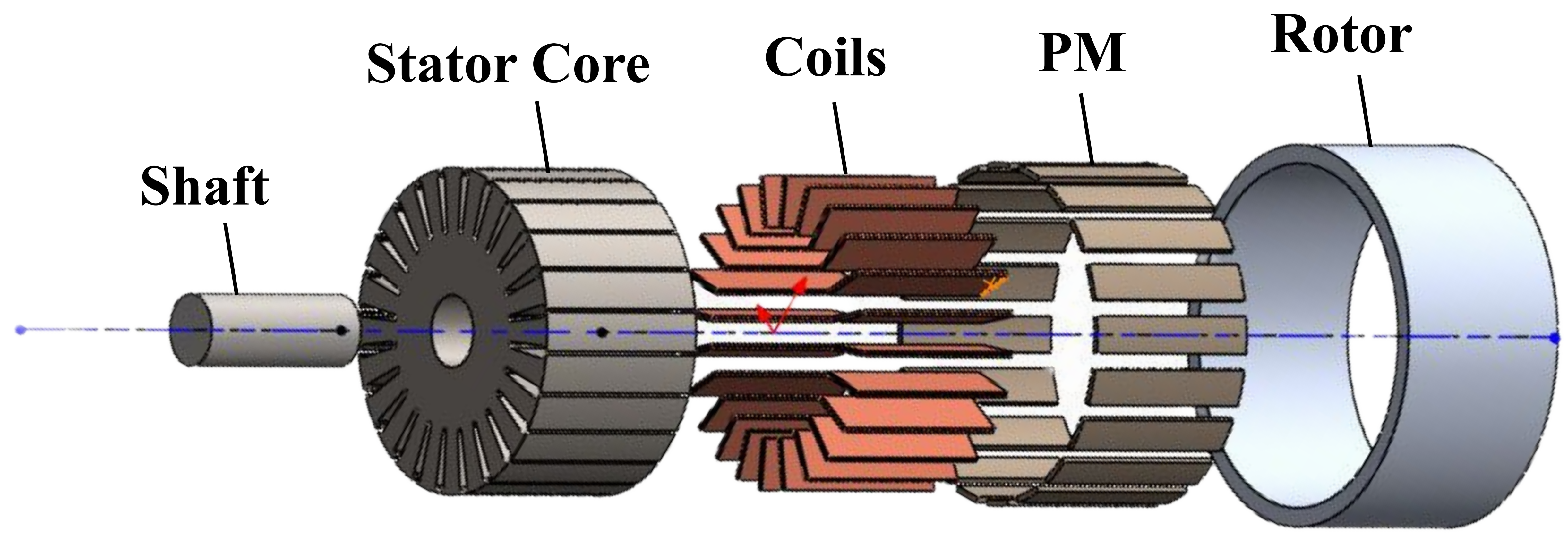
2 ***Decision Matrix***
- Induction & SRG eliminated - poor low-RPM efficiency and high torque ripple.
- AFPMG eliminated - complex manufacturing and poor cogging mitigation.

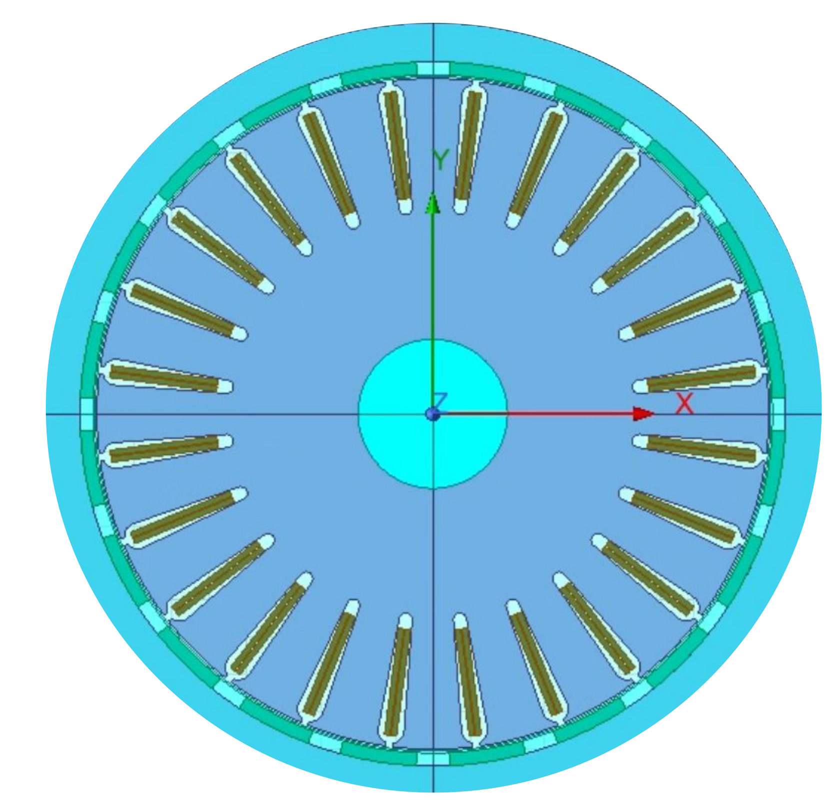
3 ***Optimization Tables - Morph Chart***
- Swept slot/pole, arc ratio, air gap, and pole segmentation

4 ***FEA Validation***
- Developed an initial analytical model using ANSYS RMxprt
- Validated electromagnetic performance in Maxwell 2D

Design Overview

Direct-drive Outer-Rotor Radial-Flux Permanent Magnet Synchronous Generator (RF-PMSG)





Key Geometry

- ***Slot / Pole***: 24/20
- ***Axial Length:*** 100mm
- ***Air Gap:*** ~1.0mm baseline
- ***Rotor OD:*** ~210 mm
- ***Winding:*** 3-Phase fractional slot concentrated winding (FSCW)
- ***Magnets***: Segmented surface mounted NdFe35

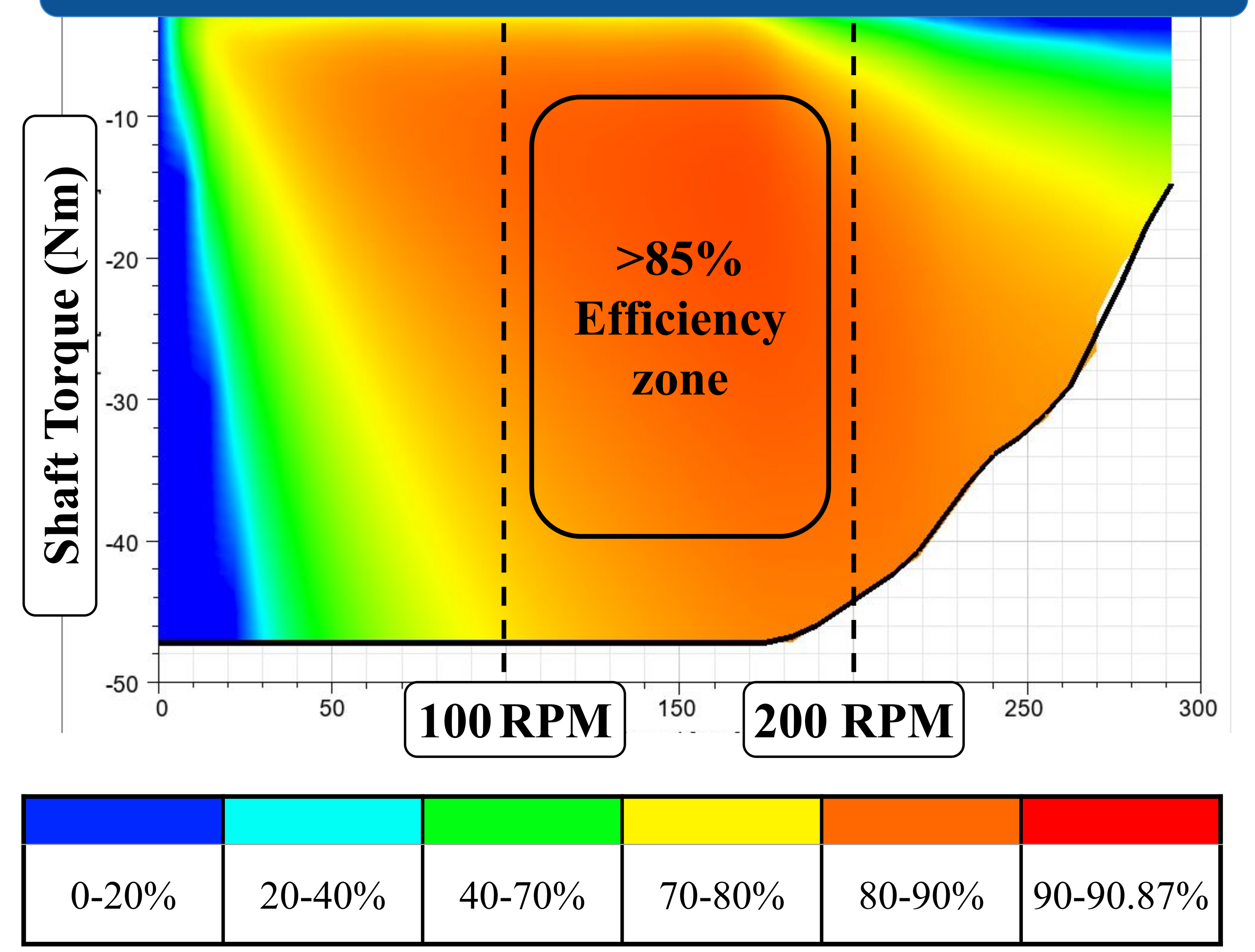
Design Goals

0.1 Nm Starting Torque	**600-1200W** Power Output
100-200 RPM Operating Speed	**>90%** Efficiency

Performance

0.09 Nm Peak Starting Torque	**842.8 W** Net Output Power
100-200 RPM Operating Speed	**67.06%** Efficiency

Efficiency Map



Next Steps

1 ***Phase 1 - Procurement***
Source NdFe35 magnets & ASTM A677 (M19) lamination steel for rotor/stator fabrication

2 ***Phase 2 - Validation***
Guided rotor-stator mating to maintain 1.0mm air gap; verify 600–1200W/0.1 N·m

MIE490/491, Capstone Design Showcase: Evaluation of Poster and Presentation

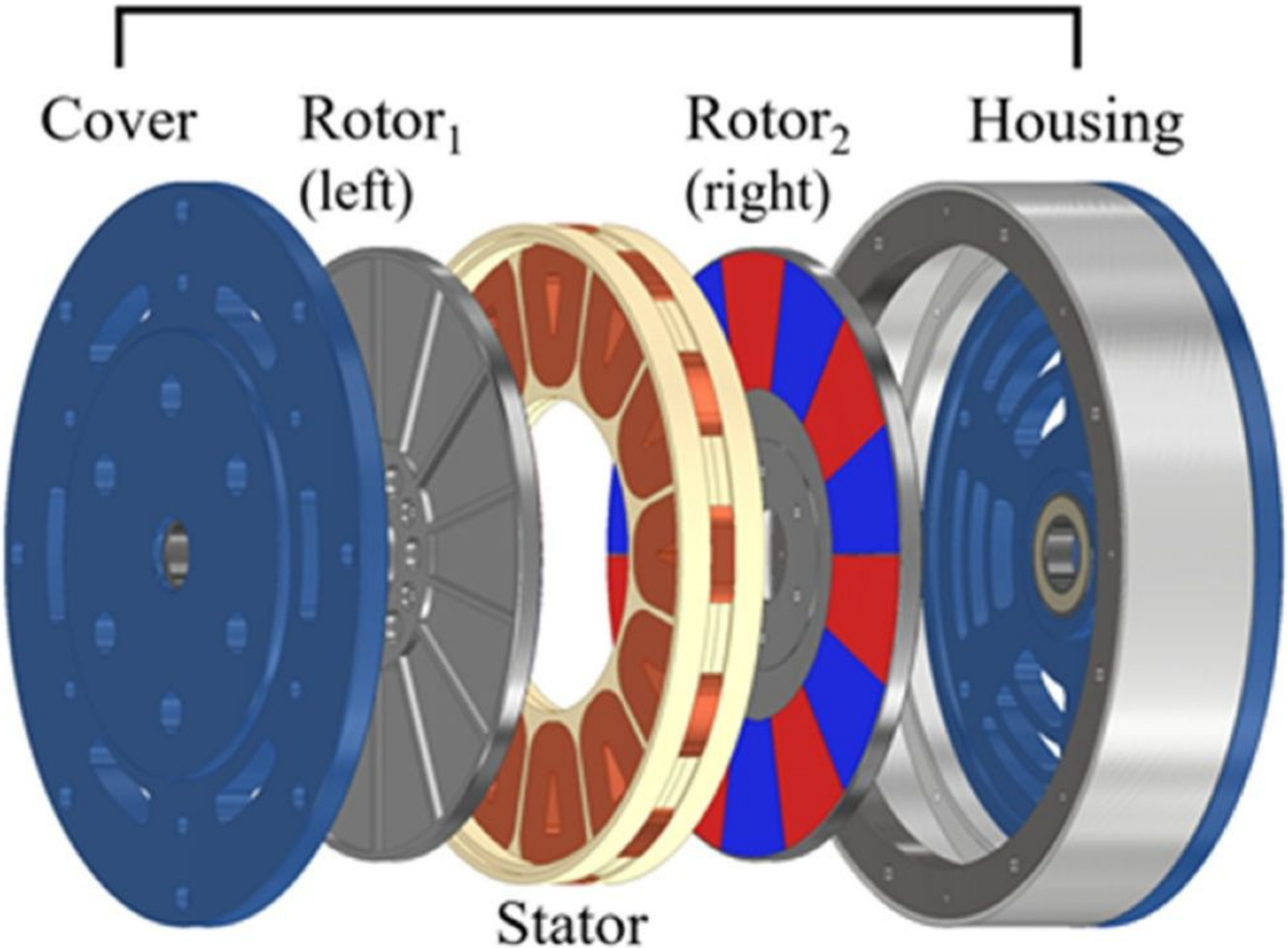
TEAM:______________ **GRADER:**________________________________

Indicator	Fails to Meet	Minimally Meets	Adequately Meets	Exceeds
Assess and provide feedback regarding individual contributions to a team activity	Severe imbalance in contributions by team members threatens meeting individual course requirements	Imbalance in contributions by team members	Balanced contributions by team members	Complementary contributions by team members
Demonstrate effective communication within the team	Effective communication is rare	Some effective communication	Regular effective communication	Constructive communication is the norm
Formulate credible and persuasive support for a claim	Lacks credible evidence and/or transitions to link evidence w/claim	Evidence may lack credibility or relevance; structure impedes linking of evidence to claim	Uses credible evidence with appropriate rhetorical tools to support claims	Presents a persuasive engineering argument in support of claim
Organize material so that the relationship of elements to a main point and to one another is clear	Lacks discernable structure	Organization is inconsistent, elements may be linked or orphaned	Relationship of elements to main point and one another clearly signaled and consistent	Structure facilitates ease of use by multiple readers with different goals
Has a logical progression of visual, textual, and oral ideas	Lacks discernable logic within and across parts of poster/presentation	Logic is inconsistent within or across parts of poster/presentation	Clear and appropriate logical progression followed within and across parts of poster/presentation	Ideas progress smoothly and logically at all levels
Tailor the mode, language, organization and content of communication to meet the needs and position of the audience(s)	Mode, language, organization and/or content inappropriate to identified audience	Inconsistent relationship between mode, language, content choices and needs of audience	Language, mode and content choices clearly indicate intended audience	Language, mode and content choices meet the needs of multiple audiences
Relate ideas in a multi-modal manner	Lacks reference to poster in presentation or presentation in poster	Inconsistent use of poster content in presentation	Supports presentation with reference to poster or vice-versa	Seamless integration of poster and presentation
Incorporate appropriate media effectively	Uses text only	Media choices ineffective or not incorporated with content	Media chosen to best accomplish a rhetorical purpose, clearly incorporated into text/presentation	Creative use of media to accomplish rhetorical purpose

Induction Generator

Switched Reluctance Generator (SRG)

Axial-Flux Permanent Magnet Generator (AF-PMG)



Outer-Rotor Radial-Flux Permanent Magnet Synchronous Generator (RF-PMSG)